PAPER ID-410435

Subject Code: KIT601

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VI) THEORY EXAMINATION 2023-24
DATA ANALYTICS

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Qno.	Question	Marks	CO
a.	List the steps involved in the analytic process.	2	1
b.	Differentiate data analysis from data reporting.	2	1
c.	Explain the concept of Bayesian inference.	2	2
d.	Define the concept of the 'kernel trick'.	2	2
e.	List the uses of sliding windows in stream processing.	2	3
f.	Explain the concept of a decaying window.	2	3
g.	Mention the importance of frequent itemsets in market basket analysis.	2	4
h.	Explain the significance of support and confidence in the context of frequent itemsets.	2	4
i.	Define Hadoop Distributed File System.	2	5
j.	List some visual data analysis techniques.	2	5

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a.	Discuss the key characteristics of data that impact data analytics.	10	1
b.	Differentiate linear and logistic regression models, and in what scenarios would each be most appropriate.	10	2
c.	Discuss differences between traditional batch data processing and stream data processing, and the main challenges associated with stream processing.	10	3
d.	Mention the limitations of the Apriori algorithm, and how they are addressed through optimization techniques.	10	4
e.	Explain the MapReduce programming model and its role in processing large datasets.	10	5

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Discuss the various phases of the data analytics lifecycle and activities involved in each phase.	10	1
b.	Explain modern data analytics tools in detail.	10	1

4. Attempt any *one* part of the following: **10 x 1 = 10**

a.	Describe the process of conducting a principal component analysis (PCA) and explain how it can be used to reduce the dimensionality of a dataset.	10	2
b.	Explain how support vector machines (SVMs) utilize kernel functions to handle non-linearly separable data.	10	2

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5. Attempt any *one* part of the following: 10 x 1 = 10

a.	Discuss a case study that demonstrates the use of stream processing in stock market prediction, including the techniques used and the results achieved.	10	3
b.	Explain Real-time Analytics Platforms (RTAP), and how do they enable real-time decision-making and analytics.	10	3

6. Attempt any *one* part of the following: 10 x 1 = 10

a.	Describe the K-means clustering algorithm and the process of initializing centroids.	10	4
b.	Explain how ProCLUS (Projected Clustering) algorithm differs from CLIQUE in its approach to high-dimensional clustering.	10	4

7. Attempt any *one* part of the following: 10 x 1 = 10

a.	Compare and contrast NoSQL databases with traditional SQL databases. List four main types of NoSQL databases.	10	5
b.	Describe Apache Hive and its use in querying and managing large datasets stored in Hadoop.	10	5

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